

Experiment 5

Aim - To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

Programming in Jenkins:

Continuous Integration is a software development practice where members of a team integrate their

work frequently, usually each person integrates at least daily leading to multiple integrations per day.

Each integration is verified by an automated build (including test) to detect integration errors as

quickly as possible.” In simple way, Continuous integration (CI) is the practice of frequently building

and testing each change done to your code automatically.

Jenkins is a self-contained, open-source automation server which can be used to automate all sorts of

tasks related to building, testing, and delivering or deploying software.

Our first job will execute the shell commands. The freestyle project provides enough options and

features to build the complex jobs that you will need in your projects.

Example 1

Example 1.1: Deploying a freestyle app in Jenkins

Creating a job:

Naming the job and setting it as freestyle:

Selecting build type as “Execute shell”:

Entering a simple command for the shell execution:

Applying and saving the project configuration:

Building the project:

Console output (after building):

Example 1.2: Taking parameters through files

Contents of script example1.cmd:

Executing script example1.cmd on the terminal:

Modifying the Jenkins project to execute the script while supplying required parameters:

Console output after building the modified project:

Example 2

Example 2.1: Running a Java program under Jenkins

Creating a simple Java program:

Compiling and running the program on the terminal:

Creating a new freestyle project:

Configure new project:

Console output after building:

Example 3

Example 3.1: Parameterise build

Creating a new freestyle project:

Enabling parameterisation and adding a String parameter:

Configuring the string parameter as Fname:

Adding a choice parameter and configuring it as City with the following choices:

Creating a script which takes 2 arguments for name and city:

Configuring build steps:

Entering parameters for build:

Console output after building:

Example 3.2: Running a Java program with parameters

Creating a Java program with an input argument:

Testing the program on the terminal:

Creating a new freestyle project:

Parameterise the project by adding a string parameter as follows:

Configure the build steps:

Entering the parameter for the build:

Console output after building:

Example 5

Example 5.1: Running a Python program

Creating a simple Python script:

Running the Python script on the terminal:

Creating a new freestyle project:

Parameterising the project with a string parameter as follows:

Configuring the build steps:

Setting the parameter for the build:

Console output after building:

Conclusion: Thus, we have successfully studied Continuous Integration and installed, configured, and understood programming with Jenkins.

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The image displays two screenshots of the Jenkins web interface, showing the process of creating a new job.

Top Screenshot: Jenkins Dashboard

- Header:** Jenkins logo, search icon, and user profile icon.
- Left Sidebar:** Links for "New Item", "Build History", "Build Queue", and "Build Executor Status".
- Main Content:**
 - Welcome to Jenkins!** This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.
 - Start building your software project**
 - Create a job (+)
 - Set up a distributed build**
 - Set up an agent (📄)
 - Configure a cloud (☁️)
 - Learn more about distributed builds (📖)

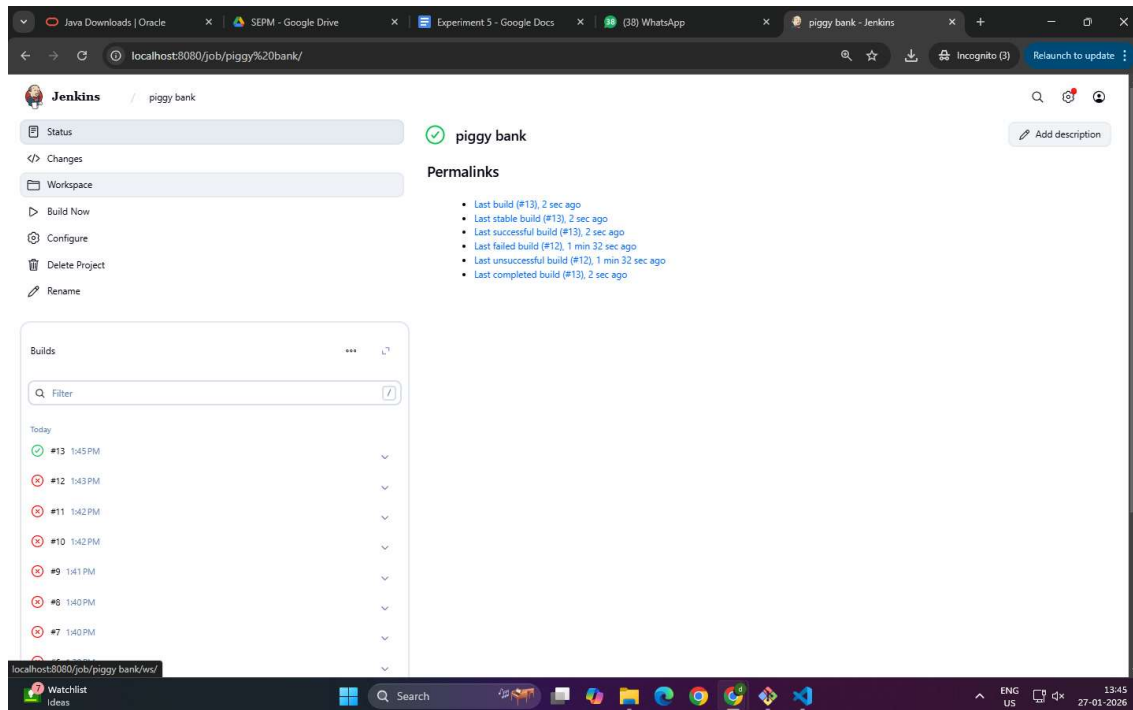
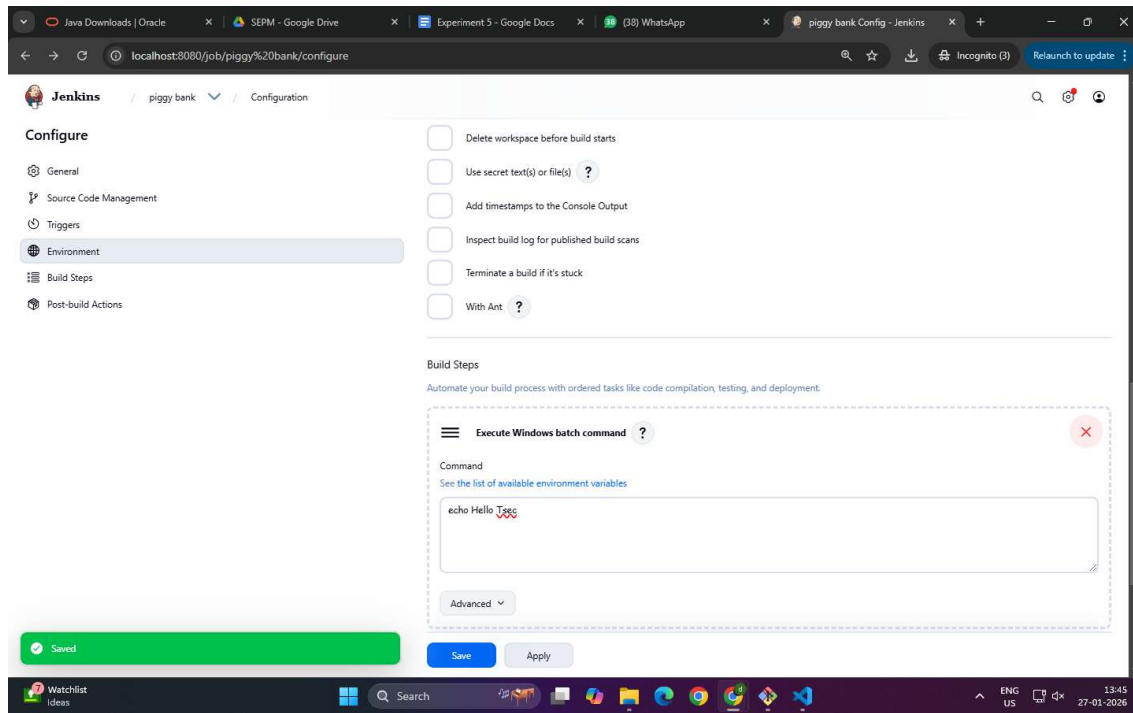
Bottom Screenshot: New Item Page

- Header:** Jenkins logo, search icon, and user profile icon.
- Left Sidebar:** Links for "New Item", "Build History", "Build Queue", and "Build Executor Status".
- Main Content:**
 - New Item** form:
 - Enter an item name:** piggy bank
 - Select an item type:**
 - Pipeline** (📄): Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
 - Freestyle project** (📄): Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
 - Multi-configuration project** (📄): Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
 - Folder** (📁): Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
 - Multibranch Pipeline** (🔑): Creates a set of Pipeline projects according to detected branches in one SCM repository.
 - OK** button

The image displays two screenshots of the Jenkins web interface, specifically the configuration page for a job named 'piggy bank'. The browser's address bar shows the URL 'localhost:8080/job/piggy%20bank/configure'.

Top Screenshot: The 'General' tab is selected in the left-hand 'Configure' menu. The 'General' section on the right shows a description field, a 'Plain text' preview, and several checkboxes: 'Discard old builds', 'GitHub project', 'This project is parameterized', 'Throttle builds', and 'Execute concurrent builds if necessary'. The 'Source Code Management' section is set to 'None'. The 'Enabled' toggle is checked. At the bottom, there are 'Save' and 'Apply' buttons.

Bottom Screenshot: The 'Build Steps' tab is selected in the left-hand 'Configure' menu. The 'Command' section shows a text area with the command 'echo "Hello Tsec"'. Below this is an 'Add build step' button. The 'Post-build Actions' section has an 'Add post-build action' button. At the bottom, there are 'Save' and 'Apply' buttons. A green 'Saved' notification bar is visible at the bottom left of the interface.



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The screenshot shows a web browser window displaying the Jenkins interface. The address bar shows the URL `localhost:8080/job/piggy%20bank/lastBuild/console`. The Jenkins logo is in the top left, followed by the job name `piggy bank` and build number `#13`. The left sidebar contains links for `Status`, `Changes`, `Console Output` (which is selected), `Edit Build Information`, `Delete build #13`, `Timings`, and `Previous Build`. The main area is titled `Console Output` with a green checkmark icon. It includes buttons for `Download`, `Copy`, and `View as plain text`. The console output text is as follows:

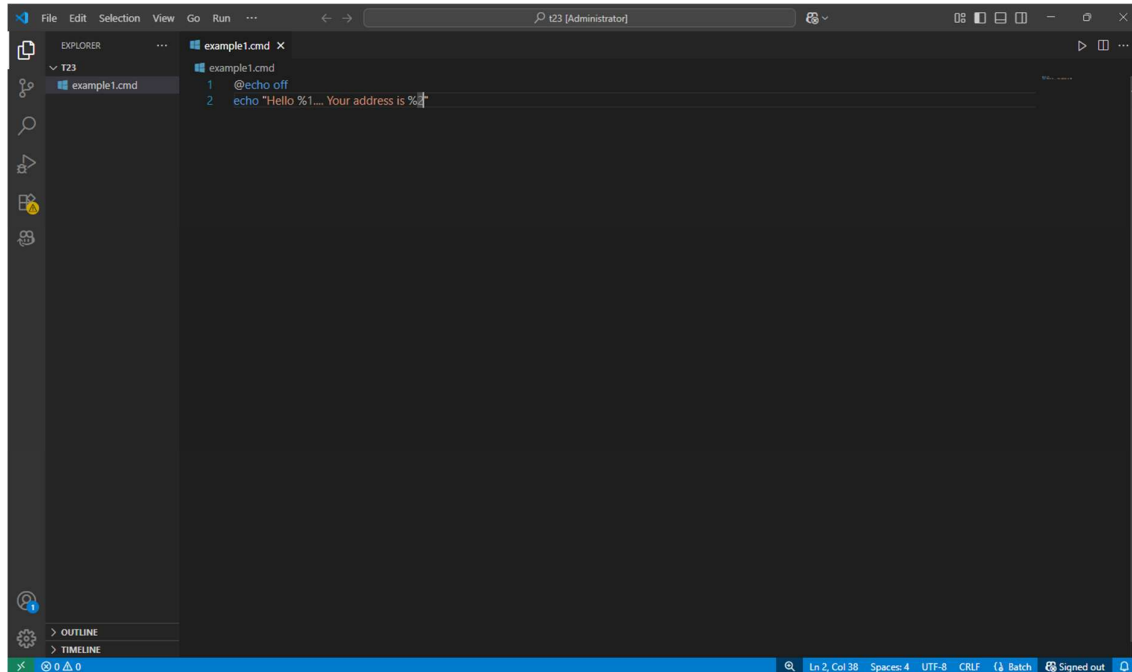
```
Started by user Tiya Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\workspace\piggy bank
[piggy bank] $ cmd /c call C:\WINDOWS\TEMP\Jenkins8989928918577008250.bat

C:\ProgramData\Jenkins\workspace\piggy bank>echo Hello Tsec
Hello Tsec

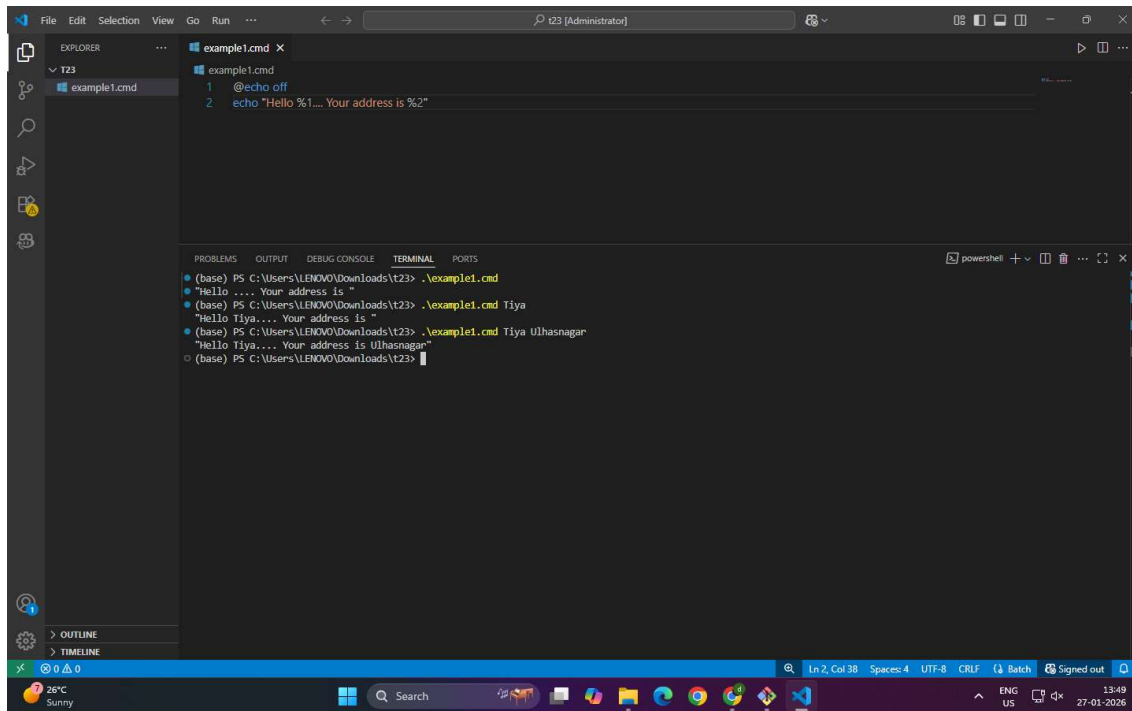
C:\ProgramData\Jenkins\workspace\piggy bank>exit 0
Finished: SUCCESS
```

At the bottom right of the Jenkins interface, it says `REST API` and `Jenkins 2.528.3`. The Windows taskbar is visible at the bottom, showing the Start button, search bar, and various application icons. The system tray shows `ENG US`, `13:45`, and `27-01-2026`.

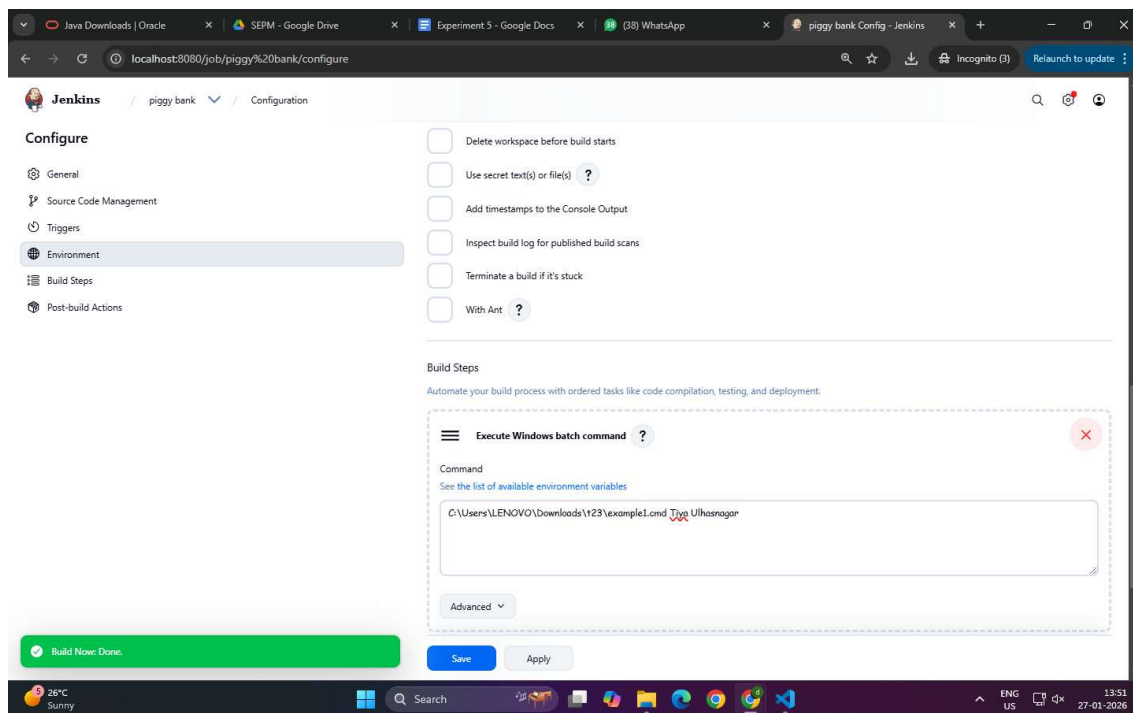
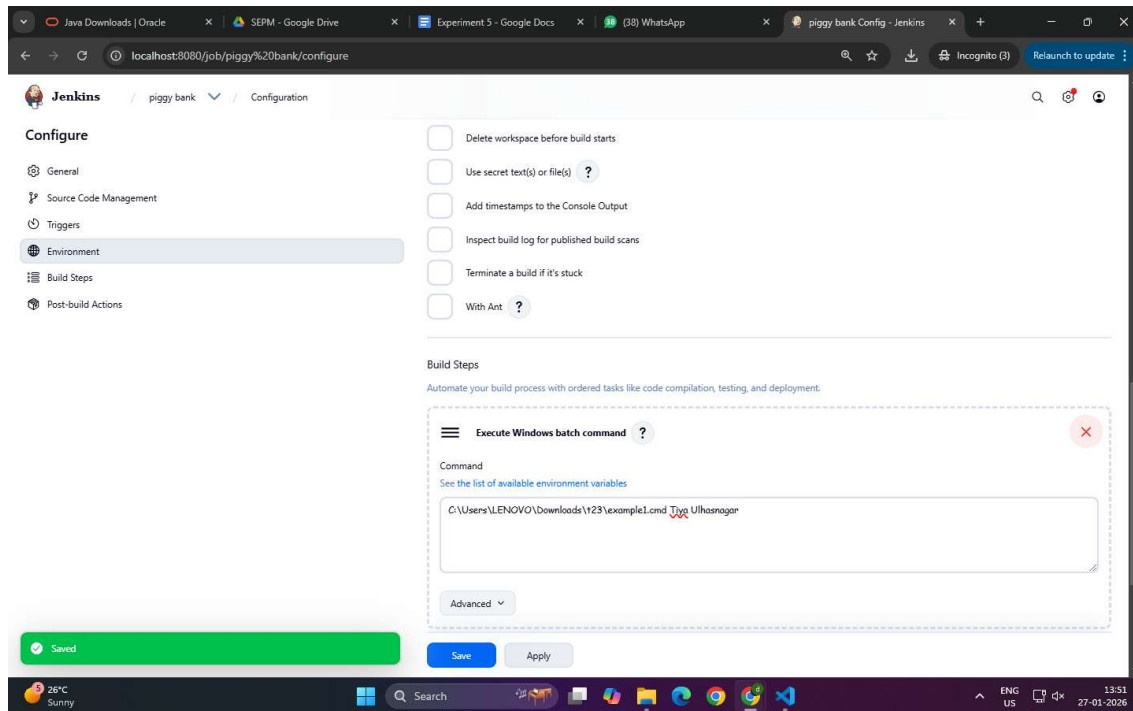
1.2

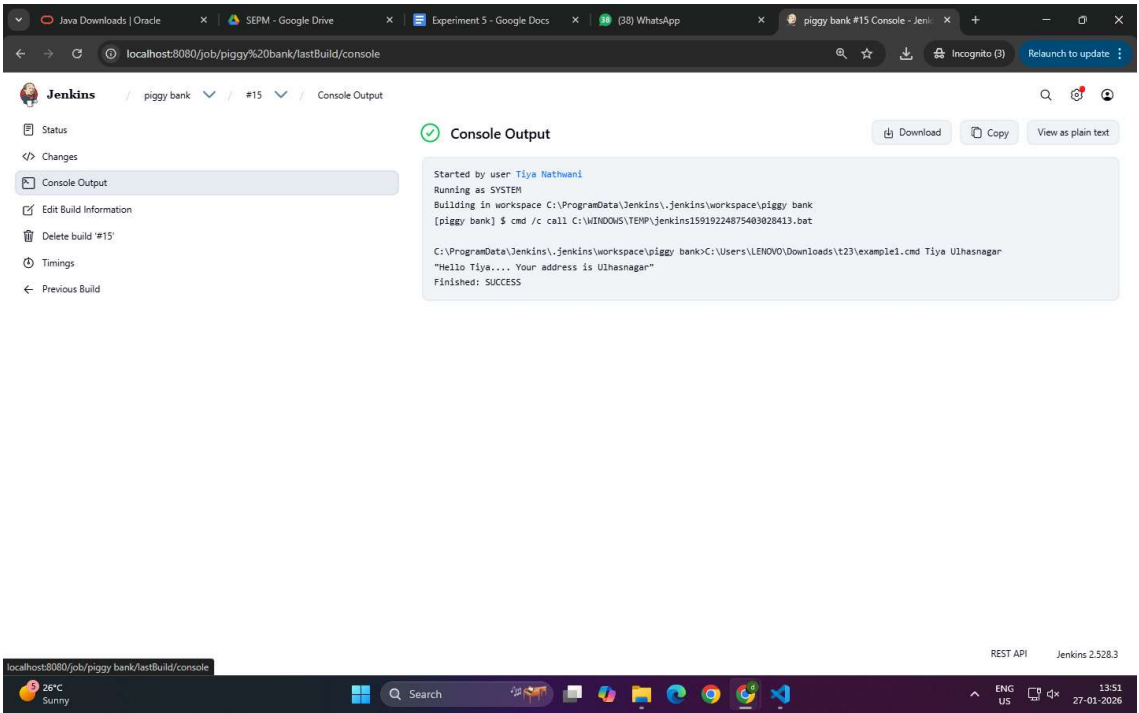


```
1 @echo off
2 echo "Hello %1.... Your address is %~"
```



```
(base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd
"Hello .... Your address is "
(base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd Tiya
"Hello Tiya.... Your address is "
(base) PS C:\Users\LENOVO\Downloads\t23> .\example1.cmd Tiya Ulhasnagar
"Hello Tiya.... Your address is Ulhasnagar"
(base) PS C:\Users\LENOVO\Downloads\t23>
```



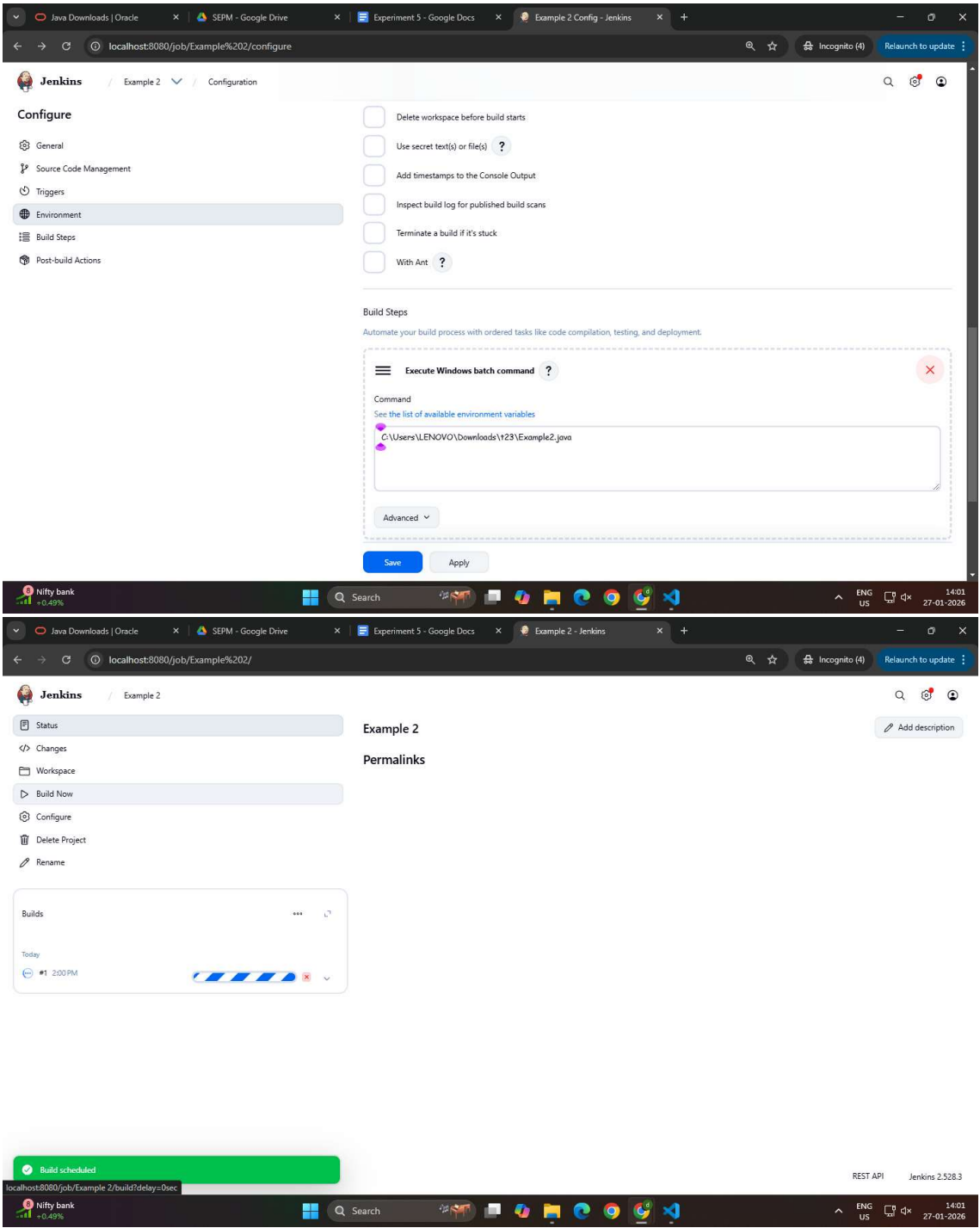


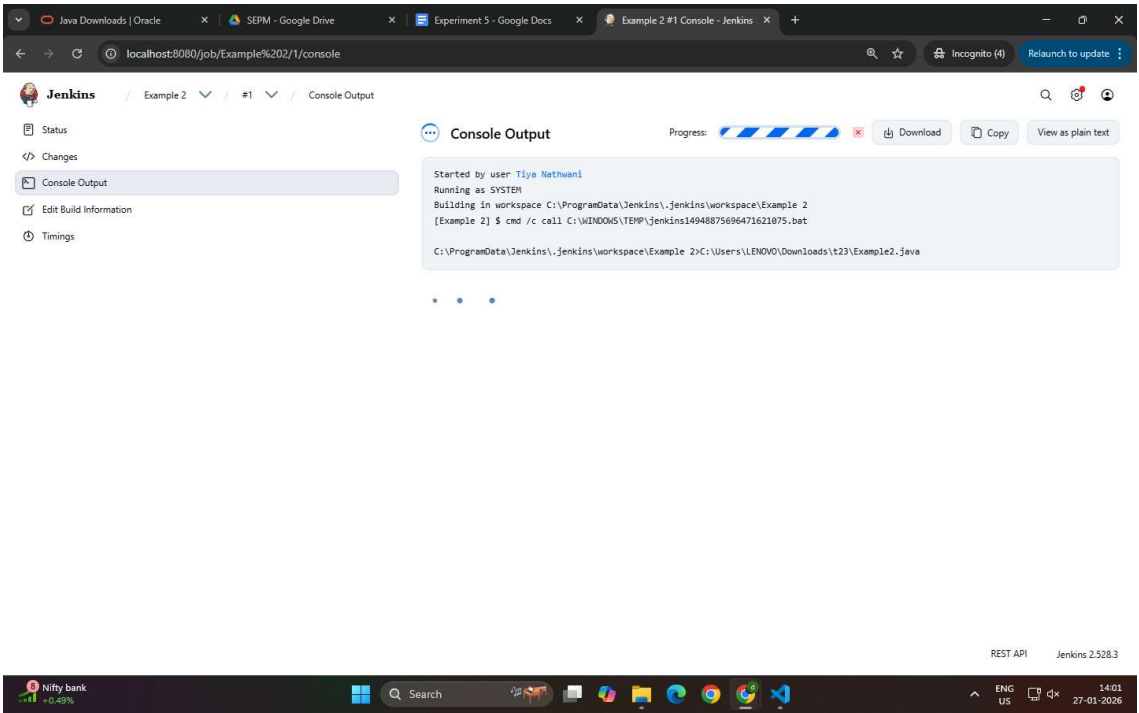
Example 2

The image shows two screenshots. The top screenshot is the Jenkins 'Configure' page for a job named 'Example 2'. The 'General' tab is selected, showing options like 'Discard old builds', 'GitHub project', 'This project is parameterized', 'Throttle builds', and 'Execute concurrent builds if necessary'. The 'Source Code Management' section is set to 'None'. The bottom screenshot is a Visual Studio Code editor showing a Java file named 'Example2.java' with the following code:

```
1 class Example2 {  
2     public static void main(String args[]) {  
3         System.out.println("Hello World!");  
4     }  
5 }
```

The terminal at the bottom shows the command prompt in a PowerShell session, indicating the current directory is 'C:\Users\LENOVO\Downloads\t23'.





Example 3

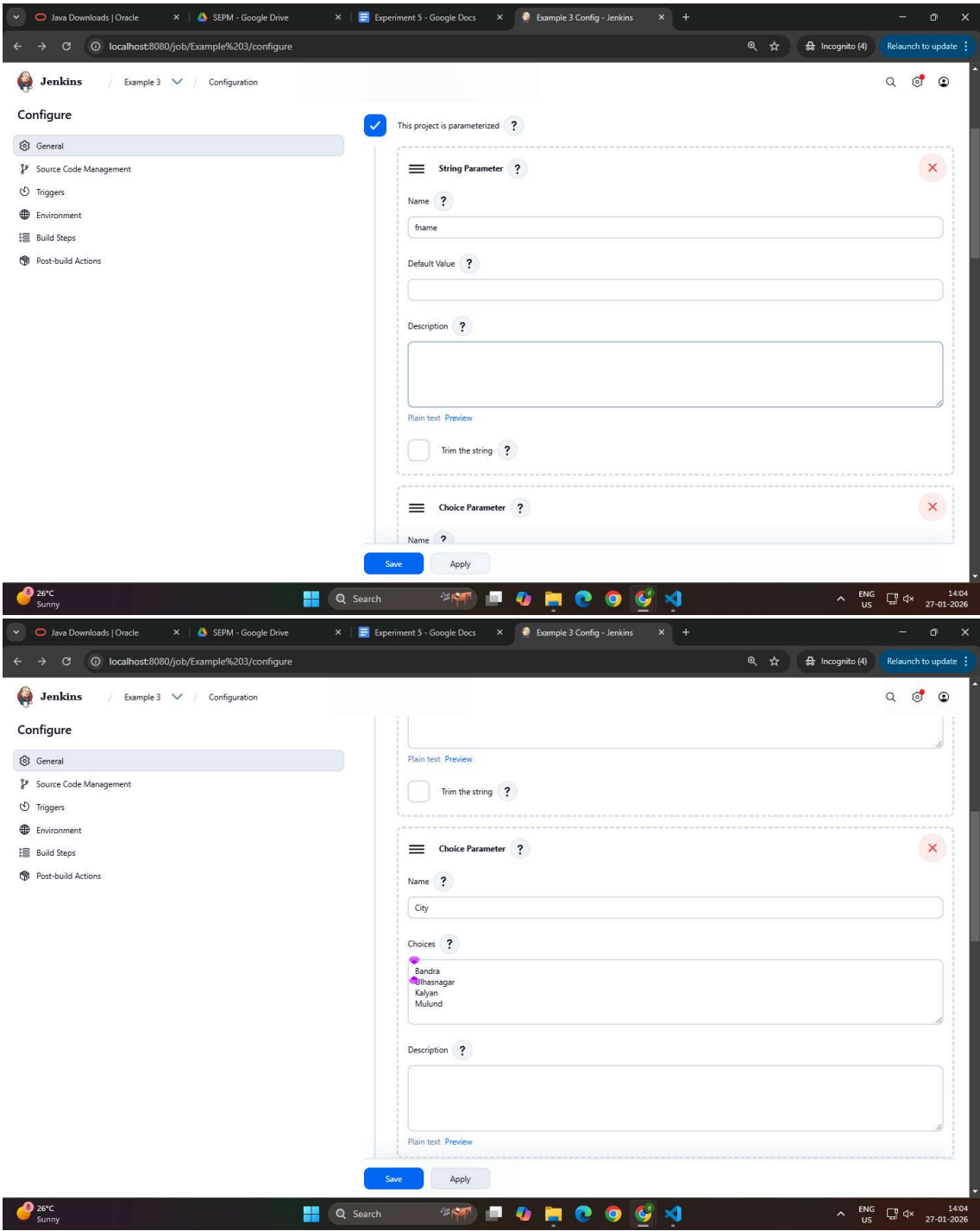
The image displays two screenshots of the Jenkins web interface, illustrating the setup of a new item.

Top Screenshot: New Item Page

- The browser address bar shows `localhost:8080/view/all/newJob`.
- The Jenkins logo and navigation links are visible at the top.
- The "New Item" section is active, with the item name "Example 3" entered in the "Enter an item name" field.
- The "Select an item type" section shows several options: Pipeline, Freestyle project (selected), Multi-configuration project, Folder, and Multibranch Pipeline.
- The "Freestyle project" description states: "Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications."
- An "OK" button is visible at the bottom.

Bottom Screenshot: Configuration Page

- The browser address bar shows `localhost:8080/job/Example%203/configure`.
- The "Configure" page is displayed, with the "General" tab selected in the left sidebar.
- The "General" tab contains the following settings:
 - ☐ GitHub project
 - ☒ This project is parameterized ?
 - ☐ Add Parameter (dropdown menu is open, showing options: Boolean Parameter, Choice Parameter, Credentials Parameter, File Parameter, Multi-line String Parameter, Password Parameter, Run Parameter, String Parameter)
 - ☐ Filter
 - ☐ Boolean Parameter
 - ☐ Choice Parameter
 - ☐ Credentials Parameter
 - ☐ File Parameter
 - ☐ Multi-line String Parameter
 - ☐ Password Parameter
 - ☐ Run Parameter
 - ☐ String Parameter
- The "Triggers" section is visible, with the following options:
 - ☐ Trigger builds remotely (e.g., from scripts) ?
 - ☐ Build after other projects are built ?
- Buttons for "Save" and "Apply" are at the bottom.



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The image shows a dual-screen setup. The top screen displays the Jenkins web interface in a browser. The URL is `localhost:8080/job/Example%203/configure`. The page title is "Configure" for "Example 3". On the left, a sidebar lists configuration sections: General (selected), Source Code Management, Triggers, Environment, Build Steps, and Post-build Actions. The main area shows a "String Parameter" configuration. A checkbox "This project is parameterized" is checked. The parameter name is "fname", and the "Trim the string" checkbox is unchecked. A "Choice Parameter" section is partially visible below. A green "Saved" notification bar is at the bottom left of the Jenkins window. The bottom screen shows the Visual Studio Code editor. The Explorer sidebar on the left shows a file tree for a project named "t23", containing `example1.cmd`, `example2.class`, `example2.java`, `example3.cmd`, and `Main.class`. The editor window is open to `example3.cmd`, showing the following content:

```
1 @echo off
2 echo "Hello %1.... Your city is %2"
```

The bottom status bar of VS Code indicates "Ln 2, Col 29", "Spaces: 4", "UTF-8", "CRLF", "Batch", and "Signed out". The Windows taskbar at the very bottom shows the system clock as 14:04 on 27-01-2026.

Jenkins

Example 3

Configuration

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command

Command

C:\Users\LENOVO\Downloads\123\example3.cmd

Advanced

+ Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

+ Add post-build action

Save

Apply

REST API

Jenkins 2.528.3

26°C Sunny

Search

ENG US

14:07 27-01-2026

Jenkins

Example 3

Status

Changes

Workspace

Build with Parameters

Configure

Delete Project

Rename

Builds

No builds

Project Example 3

This build requires parameters:

fname

tjs

City

Bandra

Build

Cancel

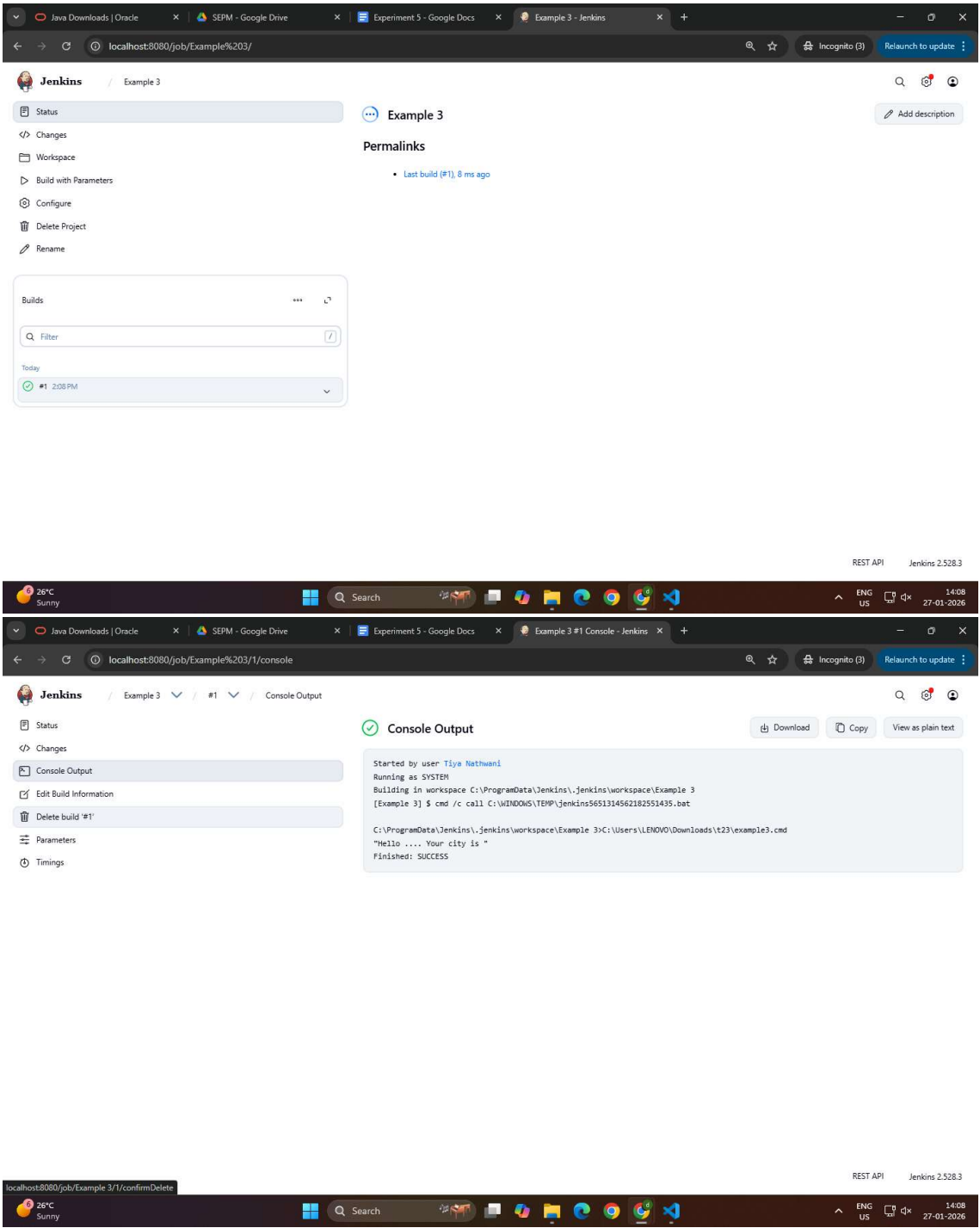
Jenkins 2.528.3

26°C Sunny

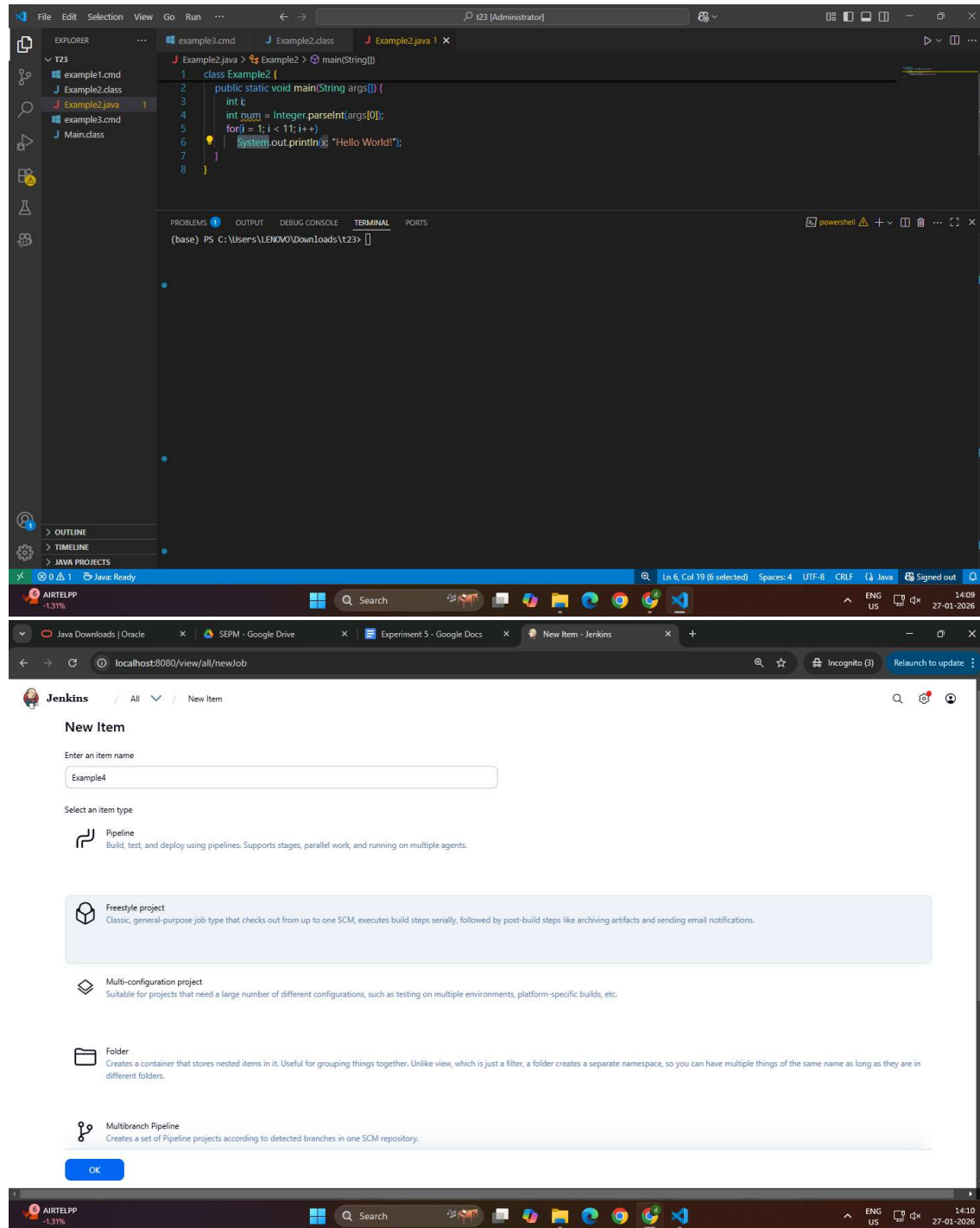
Search

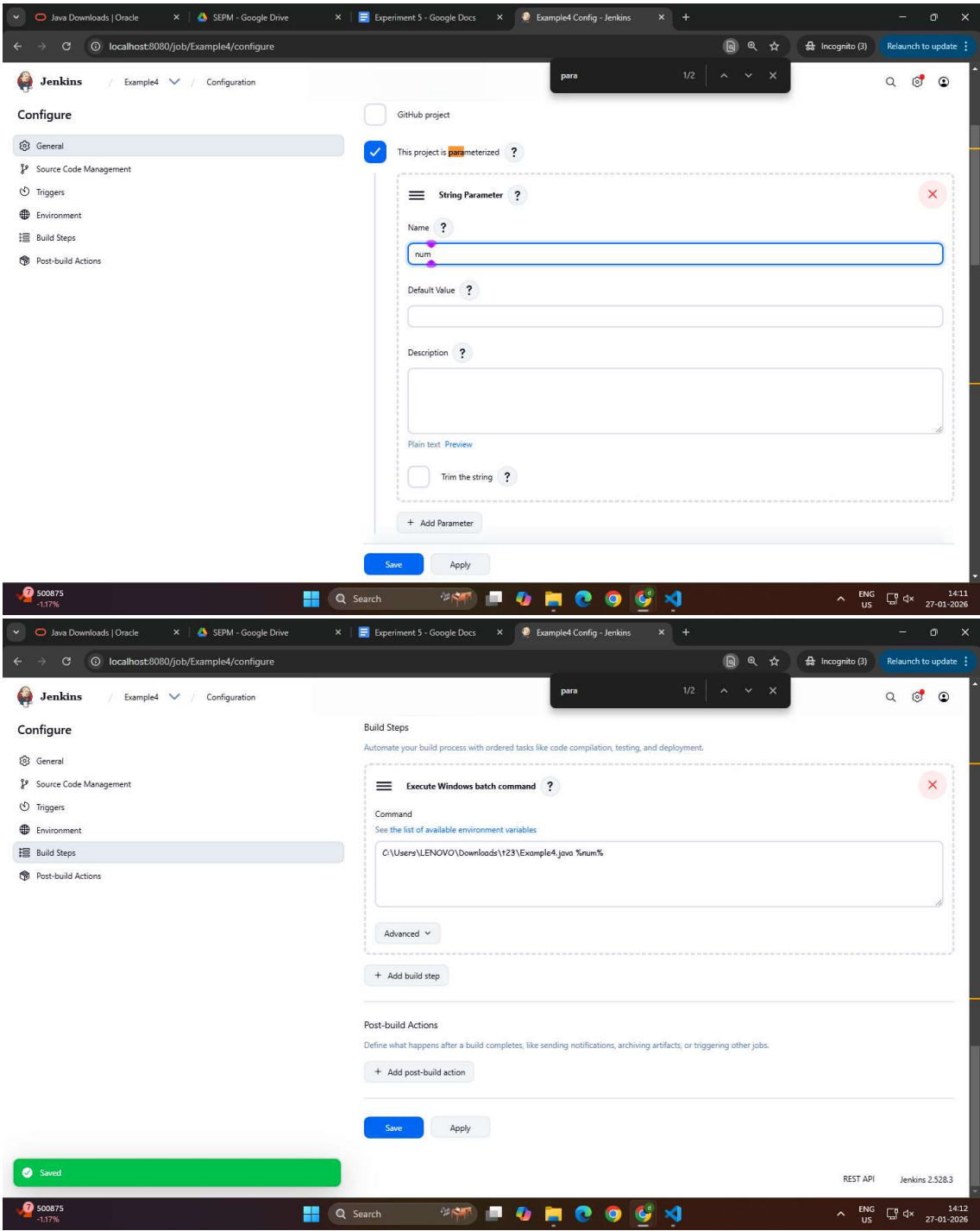
ENG US

14:07 27-01-2026

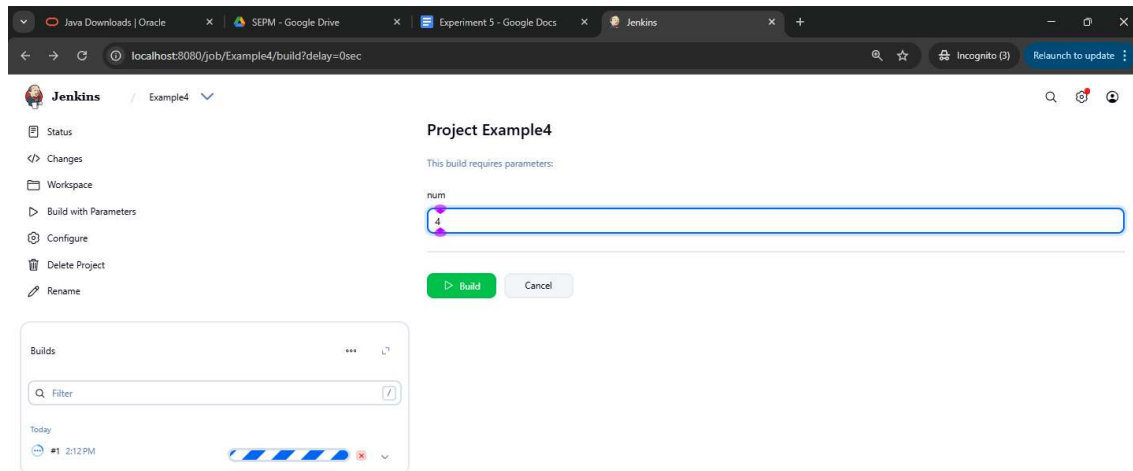


3.2

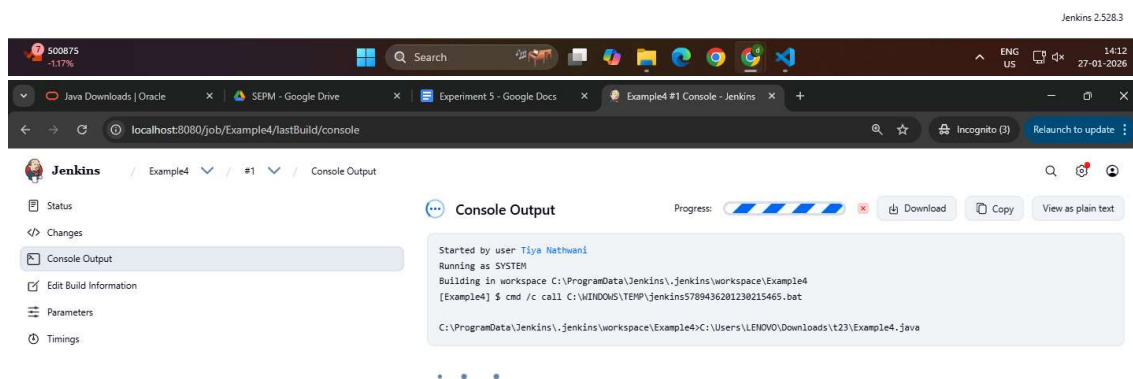




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This screenshot shows the Jenkins web interface for a project named 'Example4'. The left sidebar contains navigation links: Status, Changes, Workspace, Build with Parameters, Configure, Delete Project, and Rename. The main area is titled 'Project Example4' and indicates that the build requires parameters. A parameter named 'num' is shown with a value of '4' entered in a text field. Below the field are 'Build' and 'Cancel' buttons. On the left, a 'Builds' section shows a list of builds with a filter input and a table header 'Today'. The first build is listed as '#1 2:12 PM' with a status icon.



This screenshot shows the Jenkins web interface for the 'Console Output' of build '#1' of 'Project Example4'. The left sidebar includes links for Status, Changes, Console Output, Edit Build Information, Parameters, and Timings. The main area is titled 'Console Output' and shows the build progress as a blue bar with a red 'x' icon. The console output text is as follows:
Started by user Tiya Nathwani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example4
[Example4] \$ cmd /c call C:\WINDOWS\TEMP\jenkins5789436281238215465.bat
C:\ProgramData\Jenkins\jenkins\workspace\Example4\C:\Users\LEHDO\Downloads\t23\Example4.java
The interface also includes 'Download', 'Copy', and 'View as plain text' buttons.



This screenshot shows the Windows taskbar at the bottom of the screen. It includes the Start button, a search bar, and several pinned application icons: File Explorer, Microsoft Edge, Google Chrome, and Visual Studio Code. The system tray on the right shows the date and time as '27-01-2026 14:12' and the language as 'ENG US'.